DIGITAL LOGIC DESIGN

Course Code: CS-121

Credit Hours: **3(2+1)**

Pre-requisite: Nil

Course Objectives

The key objectives of this course include:

- Concepts and techniques underlying the construction of digitalsystems and course introduces the concept of digital logic, gates and the digital circuits it focuses on the design and analysis combinational and sequential circuits.
- It also serves to familiarize the student with the logic design of basic computer hardware components.

Course Learning Outcomes

After successful completion of this course, the students should be able to:	GA	BT Level
1. **Identify and explain** fundamental concepts of digital logic design including basic and universal gates, number systems, binary coded systems, basic concepts of digital system.	1	C3
2. Acquired knowledge to apply techniques related to the design and analyze of digital electronic circuits including Boolean algebra and multi-variable Karnaugh map methods	2	C2
3. **Model** combinational and sequential circuits using a variety of techniques & Analyze them	4	C4

Course Contents

Number Systems, Logic Gates, Boolean Algebra, Combination logic circuits and designs, Simplification Methods (K-Map, Quinn Mc-Cluskey method), Flip Flops and Latches, Asynchronous and Synchronous circuits, Counters, Shift Registers, Counters, Triggered devices & its types. Mealy machines and Moore machines. Binary Arithmetic and Arithmetic Circuits, Memory Elements, State Machines. Introduction Programmable Logic Devices (CPLD, FPGA) Lab Assignments using tools such as Verilog HDL/VHDL, MultiSim.

Mapping of CLOs to GA's

GA's/CLOs	**CLO1**	**CLO2**	**CLO3**
GA1 (Academic Education)	✓		
GA2 (Knowledge for Solving Computing Problems)		✓	
GA3 (Problem Analysis)			
GA4 (Design/Development of Solutions)			✓
GA5 (Modern Tool Usage)			
GA6 (Individual and Teamwork)			
GA7 (Communication)			
GA8 (Computing Professionalism and Society)			
GA9 (Ethics)			
GA10 (Life-long Learning)			

Resources
Text Book: 1. Digital Design. With an Introduction to the Verilog HDL by M. Morris Mano, 6th EDITION, 2021 **Reference Books** 1. Digital Logic & Computer Design by M. Morris Mano, 4^{th} edition, 2023 2. Digital Fundamentals by Floyd, 11^{th} Edition, 2021. 3. Fundamental of Digital Logic with Verilog Design, by Stephen Brown, 3^{rd} Edition, 2014

Weekly Course Plan	
Week No	**Topics**
WEEK 01	• **Digital Systems and Binary Numbers** Digital and analog systems, Digital Number Systems: Analog quantities and physical quantities the difference , how analog signal is stored, future is digital Positional and Non Positional numbers system Decimal, Binary, Octal, and Hexadecimal, Number-base Conversions
WEEK 02	• 1's and 2's complement. Subtraction using 2's complement • Signed Number representation • BCD Code, Excess-3 code, Gray Code
WEEK 03	Ch02: Boolean Logic. Boolean algebra, Boolean postulates & theorems. Expression of Digital Function in Boolean Algebra. Canonical Forms;Sum of minterms, Product of maxterms.
WEEK 04	Standard Forms: SOP, POS.
WEEK 05	Ch03. Gate Level Minimization. K-map 2 variable, 3 variable, 4 variable,5 variable, Don't Careconditions
WEEK 06	Ch04: Combinational Circuits, Design. Analysis. Half Adder, Full Adder. Half & full Subtractor, Example. BCD toExcess-3 Converter
WEEK 07	4 bit parallel binary adder, BCD Adder How to display on Seven Segment? BCD to excess 3 code and vice versa
WEEK 08	Combinational circuit building blocks. Decoder / DE multiplexer synthesis of logic functions using decoder, multiplexer / DE multiplexer. Function implementation using multiplexers

MID SEMESTER EXAMINATION	
WEEK 9& 10	Ch05: Sequential Circuits. Introduction. Latches: SR, S'R', D Latch. Edge Triggering. Negative Edge triggered D Flip-flop
WEEK 11	Positive-Edge triggered D Flip-flop JK Flip Flop, T Flip Flop: Characteristic Table, characteristic equations
WEEK 12	Analysis of Clocked Sequential Circuits. State Equation, State Table, State Diagram using D Flip-Flop Analysis with JK Flip Flop. Examples. Analysis with T Flip Flop. Examples
WEEK 13	State Reduction and Assignments. Design of clocked Sequential Circuits.Design with D Flip Flop
WEEK 14	Design with JK Flip Flop. Design with T Flip Flop.Examples
WEEK 15	Ch 06: Registers. 4-bit registers. Shift registers.Serial addition, Universal Shift Register
WEEK 16	Synchronous counters: Binary counter Up Down binary counter BCD counter. Binary counter with Parallel load Ring Counters, Johnson Counter State Machines. Introduction Programmable Logic Devices (CPLD, FPGA)
END SEMESTER EXAMS	

DIGITAL LOGIC DESIGN LAB

LAB Objectives

The key objectives of this LAB include:

1. Able to design circuits on Trainer and on MultiSIM.
2. Understand Boolean algebra and basic properties of Boolean algebra; able to simplify simple Boolean functions by using the basic Boolean properties and implement them practically.
3. Able to design simple combinational logics using basic gates. Able to optimize simple logic using Karnaugh maps, understand "don't care" and implement them on Trainers.
4. Familiarize students with Lab apparatus such as Wires, ICs, Trainers, LEDs and encourage students to create projects.
5. Familiarize students with basic combinational and sequential components used in the typical data path designs: Register, Adders, Shifters, Comparators; Counters with implementation in MultiSIM and Trainers.
6. Able to understand Verilog HDL/VHDL simulation

Lab Learning Outcomes

After successful completion of this course, the students should be able to:	GA	BT Level
1. **Acquire knowledge** related to the concepts, tools and techniques for the design of digital electronic circuits	1	C2
2. **Demonstrate the** skills to design and analyze both combinational and sequential circuits using a variety of techniques	4	P4
3. P **erform the analysis of** small-scale combinational and sequential digital circuits	3	P5
4. **Practice** the behavior of digital circuits using modern tools	5	P3

LAB Contents

Familiarization and verification of basic logic gates using ICs, Verification of basic identities of Boolean Algebra, Verification of De Morgan's law, Verification of XOR and XNOR Circuitry, Introduction to Verilog, Implementation in SOP and POS forms, Implementation of full and half adders, Implementation of 2-4 decoder and 3-8 decoder, Implementation of 4-input priority encoder, Implementation of BCD to seven segment converter, Implementation of half adder and full adder in Verilog, Implementation of BCD to excess-3 code conversion, Implementation of Multiplexer and Demultiplexer with RS-Flip flop via NAND Gates, Implementation of 4-bit asynchronous counter, implementation of 3-bit Ripple Counter, Implementation of Ring Counter with introduction to programmable devices.

Resources

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Mapping of LLOs to GA's

GA's/LLOs	LLO1	LLO2	LLO3	LLO4
GA1 (Academic Education)	✓	✓		
GA2 (Knowledge for Solving Computing Problems)				
GA3 (Problem Analysis)			✓	
GA4 (Design/Development of Solutions)				✓
GA5 (Modern Tool Usage)				
GA6 (Individual and Teamwork)				
GA7 (Communication)				
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Weekly List of Experiments

Week no.	Topics
1	Familiarization and verification of basic logic gates using ICs.
2	Verification of basic identities of Boolean algebra: • Identity Law • Null (or Dominance) Law • Idempotent Law • Inverse Law • Double Complement Law • Associative Property • Distributive Property
3	(a) Verification of DE Morgan's Law (b) Development and Verification of XOR and XNOR gates using universal gates

4	**Implementation** of the given Boolean function using logic gates in both SOP and POS forms
5	**Implementation** of half adder and full adder (a) Using basic gates (b) Using universal gates
6	**(a) Implementation and verification** of Half adder in Verilog **(b) Implementation and verification** of Full adder in Verilog
	(a) **Implementation** of 2-to-4 decoder and 3-to-8 line decoder with enable input **(b) Implementation** of 4-input priority encoder
7	**Implementation** of BCD to seven segment converter & 4 x 1 multiplexer and 1 x 4 DE multiplexer circuit.
8	**Lab Sessional**
9	Verification of Latch and Flip Flop operation using gates and flip flop's IC
10	**Implementation** of RS flip-flop and D flip-flop using NAND gates.
11	**Implementation** of series and parallel registers
12	**Implementation** of 4-bit binary asynchronous counter
13	**Implementation** of asynchronous (3-bit Ripple) Counter.
14	**Implementation** of Ring Counter and Johnson Counter.
15	**Implementation** of RAM and ROM using gates and Static RAM IC
16	**Lab Sessional**